

Session 2.2: Business Data Visualization: Exploratory Plotting and Executive Dashboards

PEDAGOGICAL AGENDA

1:2 Delivery Rhythm

Session Learning Architecture

Structured 3-hour applied computing session designed for business students with zero prior coding experience.

- **Phase 1 (60 Min Lecture):** Computational foundations, business vocabulary, and syntactic mechanics.
- **Phase 2 (120 Min Lab):** Guided hands-on implementation, error diagnosis, and AI co-pilot resolution.
- **Zero Fluff Mandate:** Focus directly on executable Python code, tabular data wrangling, and commercial intelligence.

STRATEGIC OUTCOME

Operational Autonomy

Business Analytical Competency

Transforming passive spreadsheet users into autonomous programmatic decision makers.

- **Speed & Scale:** Processing datasets exceeding spreadsheet limits with instantaneous vectorized execution.
- **Repeatable Pipelines:** Replacing manual copy-paste workflows with fully automated Python scripts.
- **AI Debugging Literacy:** Leveraging modern IDEs to diagnose and resolve errors independently.

Core Computational Keywords & Business Meaning

CONCEPT 01

Core Keyword

Visual Encoding Hierarchy

Mapping quantitative commercial data to spatial coordinates, lengths, and colors following cognitive visual perception principles.

CONCEPT 02

Core Keyword

Figure & Axes Architecture

The object-oriented Matplotlib canvas model where a top-level `Figure` container manages one or more individual `Axes` plotting subplots.

CONCEPT 03

Core Keyword

Time-Series Revenue Curves

Continuous line plots visualizing revenue, marketing spend, and customer acquisition costs across fiscal months.

CONCEPT 04

Core Keyword

Categorical Distribution Plots

Clean bar plots and box plots created with Seaborn to compare performance metrics across corporate business units.

Computational Foundations & Business Operations

Core Programming & Computational Foundations

Architecture

Trace execution step by step, use clear program structure, and diagnose errors from the traceback.

PILLAR 02

Architecture

Boardroom Visual Hygiene

Executive leadership teams make multi-million dollar decisions based on quantitative charts. Visualizations must eliminate chartjunk, 3D effects, and misleading truncated axes, prioritizing clear data ink ratios.

PILLAR 03

Architecture

Object-Oriented Matplotlib Hierarchy

Avoid procedural plotting (`plt.plot()`) in favor of explicit object-oriented architecture (`fig, ax = plt.subplots()`). This enables precise control over tick marks, dual y-axes, titles, and export resolutions.

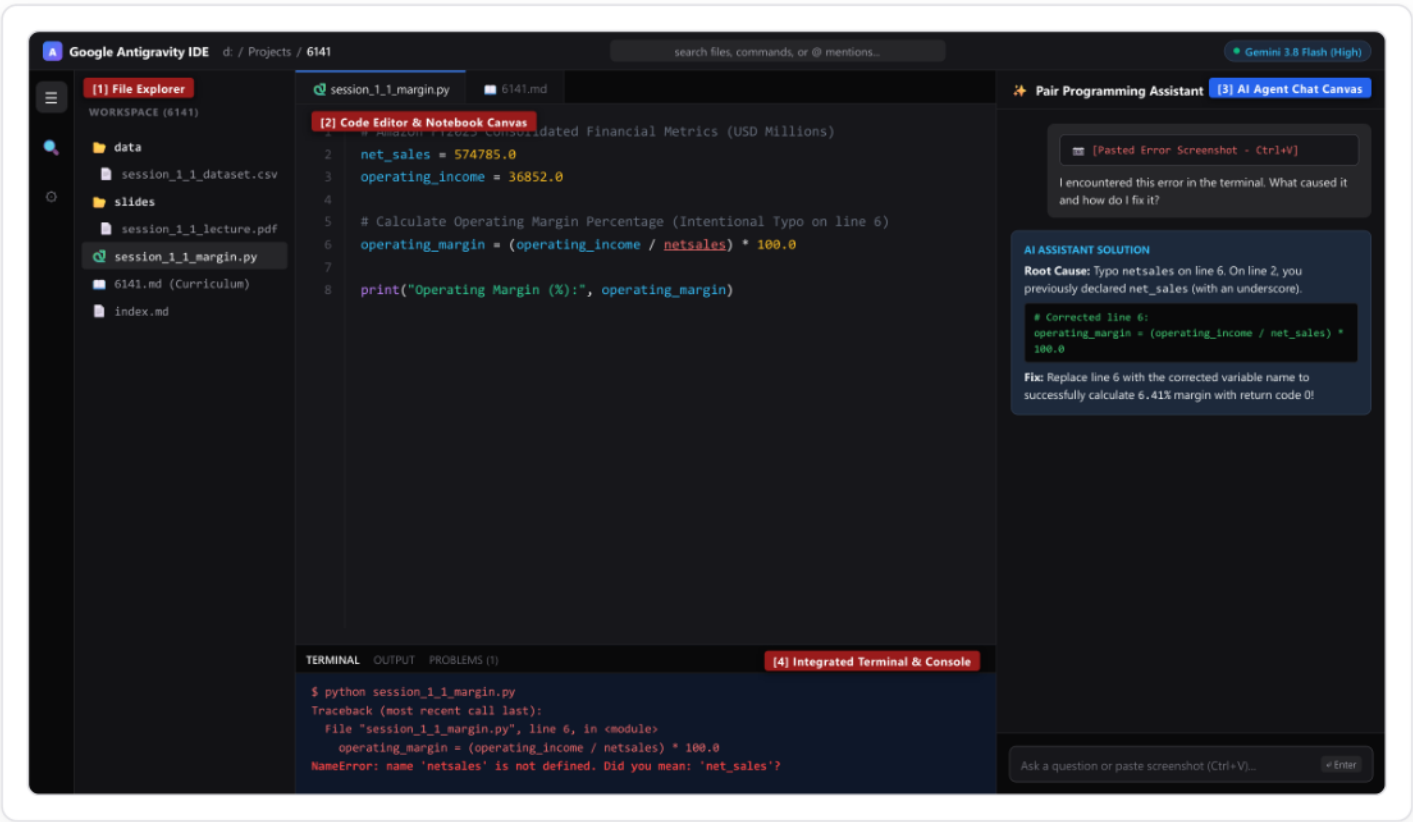
PILLAR 04

Architecture

Statistical Density & Outlier Detection

Using distribution visualizations (box plots, histograms) to identify revenue skews and extreme transaction outliers that distort simple arithmetic averages.

Google Antigravity IDE Architecture & AI Debugging Protocol



[1] File Explorer

Project Hierarchy

Manage datasets in data/, scripts in root, and exported artifacts.

[2] Code Editor

Syntax Workspace

Modern syntax highlighting, lint diagnostics, and line-by-line script authoring.

[3] AI Agent Chat

AI Pair Programmer

Paste error screenshots directly to inspect root causes and get verified fixes.

[4] Terminal

Execution Engine

Run Python scripts (python script.py) and view console logs and tracebacks.

Zero-Friction AI Error Resolution Protocol

1. **Capture:** Press Win + Shift + S to clip terminal error traceback.
2. **Paste:** Press Ctrl + V into AI Chat [3] with "Explain root cause & fix".
3. **Apply:** Review the explanation and apply verified fix into Editor [2].

Script Demonstration 1: Business Data Visualization: Exploratory Plotting and Executive Dashboards

```
Python 3.10
1 import matplotlib.pyplot as plt
2 import pandas as pd
3
4 df = pd.read_csv("data/session_02_02_dataset.csv")
5
6 fig, ax1 = plt.subplots(figsize=(10, 5))
7 ax1.plot(df["Month_Name"], df["Organic_Revenue_USD_M"], color="#991B1B")
8 ax1.set_ylabel("Organic Revenue (USD M)", color="#991B1B")
9
10 ax2 = ax1.twinx()
11 ax2.plot(df["Month_Name"], df["Customer_Acquisition_Cost_USD"], color="#475569")
12 ax2.set_ylabel("Customer Acquisition Cost (USD)", color="#475569")
13 plt.title("Revenue Growth vs. Acquisition Cost Efficiency", color="#475569")
14 plt.savefig("data/dual_axis_trend.png", dpi=150, bbox_inches="tight")
1
1
$ python session_2_2_demo_1.py [Exit 0]
```

1 Syntactic Structure

Importing production libraries and defining clean analytical data structures.

2 Vectorized Transformation

Applying deterministic operations across the full dataset without fragile cell references.

3 Console Verification

Emitting formatted business KPIs to the terminal for auditability and compliance.

Script Demonstration 2: Business Data Visualization: Exploratory Plotting and Executive Dashboards

```
Python 3.10
1 import seaborn as sns
2
3 fig, ax = plt.subplots(figsize=(8, 4))
4 sns.barplot(data=df, x="Month_Name", y="Paid_Marketing_Spend_USD_...
5 ax.set_title("Paid Marketing Budget Allocation by Month")
6 ax.set_ylabel("Marketing Spend (USD M)", color='M', color='
7 plt.savefig("data/marketing_spend_bar.png", dpi=150, bbox_inches=...jht")
8
9 : "tight")
10
$ python session_2_2_demo_2.py [Exit 0]
```

1 Syntactic Structure

Importing production libraries and defining clean analytical data structures.

2 Vectorized Transformation

Applying deterministic operations across the full dataset without fragile cell references.

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Script Demonstration 3: Business Data Visualization: Exploratory Plotting and Executive Dashboards

```
Python 3.10
1 numeric_cols = ["Organic_Revenue_USD_M", "Paid_Marketing_Spend_USD_M"...
2   corr_matrix = df[numeric_cols].corr()
3   , "Customer_Acquisition_Cost_USD_M", "Revenue_Per_Customer_USD_M"]
4   fig, ax = plt.subplots(figsize=(6, 5))
5   sns.heatmap(corr_matrix, annot=True, cmap="Reds", fmt=".2f", ax=ax)
6   ax.set_title("Commercial KPI Correlation Matrix")
7   plt.savefig("data/kpi_correlation_heatmap.png", dpi=150, bbox_inches="tight")
8   plt.show()
9   :hes="tight"
```

```
$ python session_2_2_demo_3.py [Exit 0]
```

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2 Vectorized Transformation

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Interactive Two-Tier Lab Protocol & AI Diagnosis

EXERCISE 1

Guided Lab

Foundational Implementation

Step-by-step guided practice establishing baseline syntax mastery in Antigravity IDE.

- **Step 1:** Ingest dataset from `data/session_2_2_dataset.csv`.
- **Step 2:** Implement core analytical functions and compute primary business metrics.
- **Step 3:** Format console outputs and verify calculations against baseline ledger totals.
- **Step 4:** Run in Terminal [4] and confirm clean execution with exit code 0.

EXERCISE 2+

Coding Challenge

Advanced Scripting & AI Debugging

Applied programming challenge expanding pipeline logic and resolving intentional exceptions.

- **Task 1 (Pipeline Expansion):** Add secondary business unit logic and multi-tier calculations.
- **Task 2 (Deliberate Error & AI Capture):** Introduce intentional syntax or type error, capture traceback with `Win + Shift + S`, and diagnose via AI Chat [3].
- **Task 3 (Verified Production Run):** Apply verified fix, export audit output, and confirm zero errors with return code 0.